



Employee Attrition Prediction Using Machine Learning

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What is Employee Attrition?

Employees leaving an organization voluntarily or involuntarily.



Purpose

Identify employees likely to leave using **Machine Learning**.



Increased Costs

Recruitment & Training



Reduced Productivity

Lower team output



Loss of Experience

Knowledge drain



ML Solution

Predict turnover early

Problem Statement



Increased Hiring Expenses



Loss of Skilled Employees



Reduced Team Performance



Business Disruption



Proposed Solution

Develop a **Machine Learning Model** to predict employee attrition.

Identify Risk Early

Take Proactive Action



Data-Driven HR



Project Objectives



Analyze Data

Examine employee data patterns and trends



Identify Factors

Discover key drivers affecting attrition



Build Models

Develop predictive machine learning models



Improve Retention

Enhance employee retention strategies



Support HR Decision-Making

Provide data-driven insights for proactive HR interventions



Dataset Information



Employee Attrition Prediction Dataset

Source: Kaggle

Features Include:



Age



Gender



Department



Job Role



Monthly Income



Years at Company



Job Satisfaction



Attrition Status



Target Variable



No
Employee Stays



Yes
Employee Leaves

<> Technology Stack



Programming



Python

Core Language



Libraries



Pandas



NumPy



Matplotlib



Seaborn



Scikit-Learn



Dev Tools



Jupyter Notebook



VS Code



GitHub

Project Workflow



Data Collection

Gather employee datasets



Data Cleaning

Remove missing values and duplicates



Data Preprocessing

Encode and split data



Exploratory Data Analysis

Visualize patterns and insights



Feature Engineering

Select and create relevant features



Model Training

Train Random Forest Classifier



Model Evaluation

Assess accuracy and metrics



Prediction

Forecast employee attrition



Data Preprocessing



Data Cleaning

- ✓ Remove missing values
- ✓ Handle duplicates
- ✓ Outlier treatment



Feature Encoding



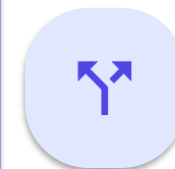
Convert categorical data to numerical



Label encoding for ordinal data



One-hot encoding for nominal data

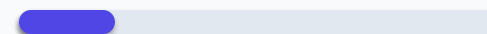


Data Splitting

Training Data **80%**



Testing Data **20%**





Exploratory Data Analysis



Visualizations



Attrition Distribution



Department-wise Attrition



Salary Distribution



Job Satisfaction Analysis

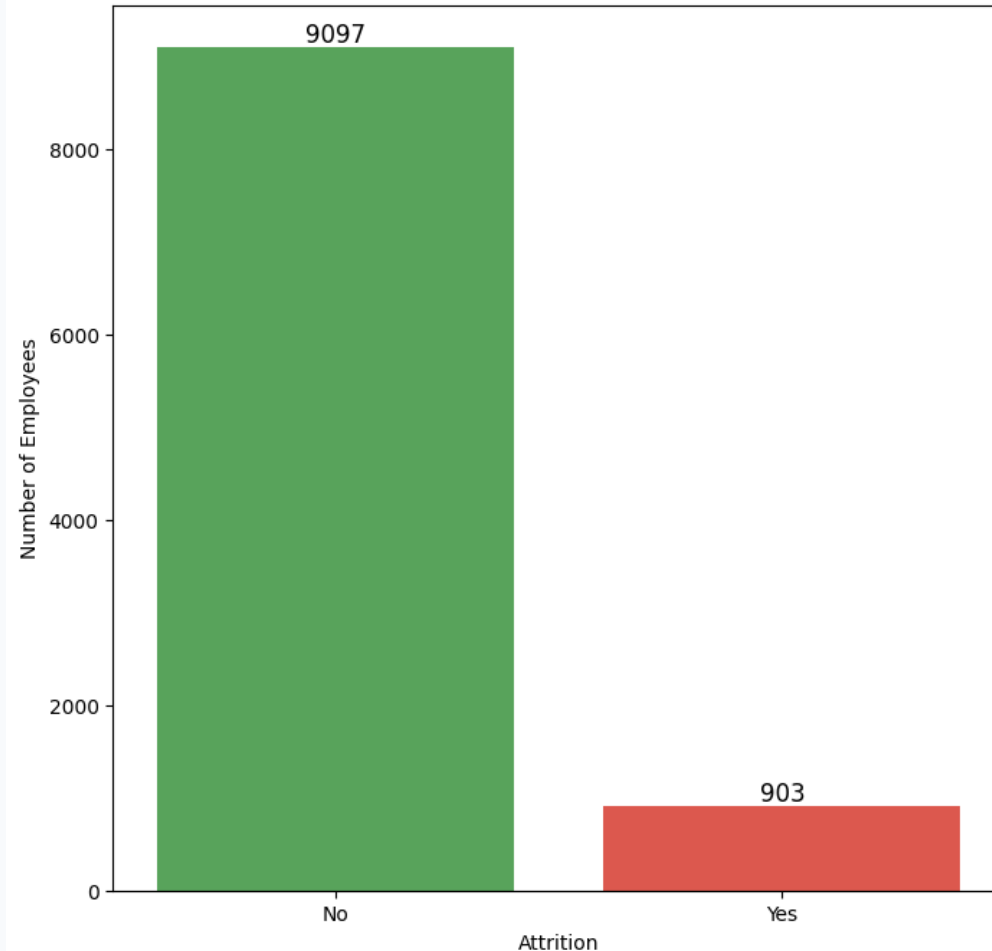


Benefits

- ✓ Understand employee behavior
- ✓ Identify important patterns
- ✓ Discover key risk factors

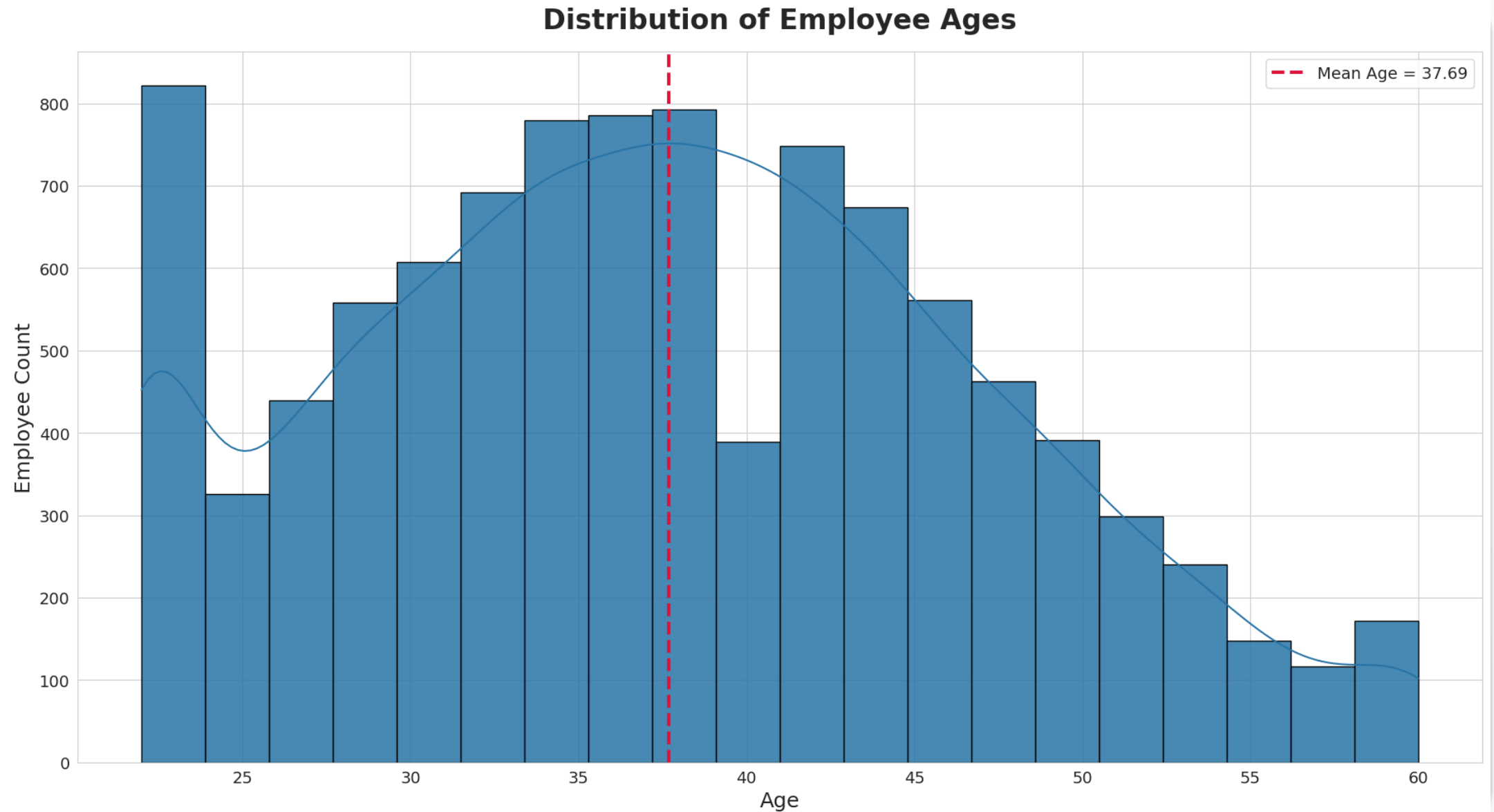
Data Analysis Graphs

Employee Attrition Distribution



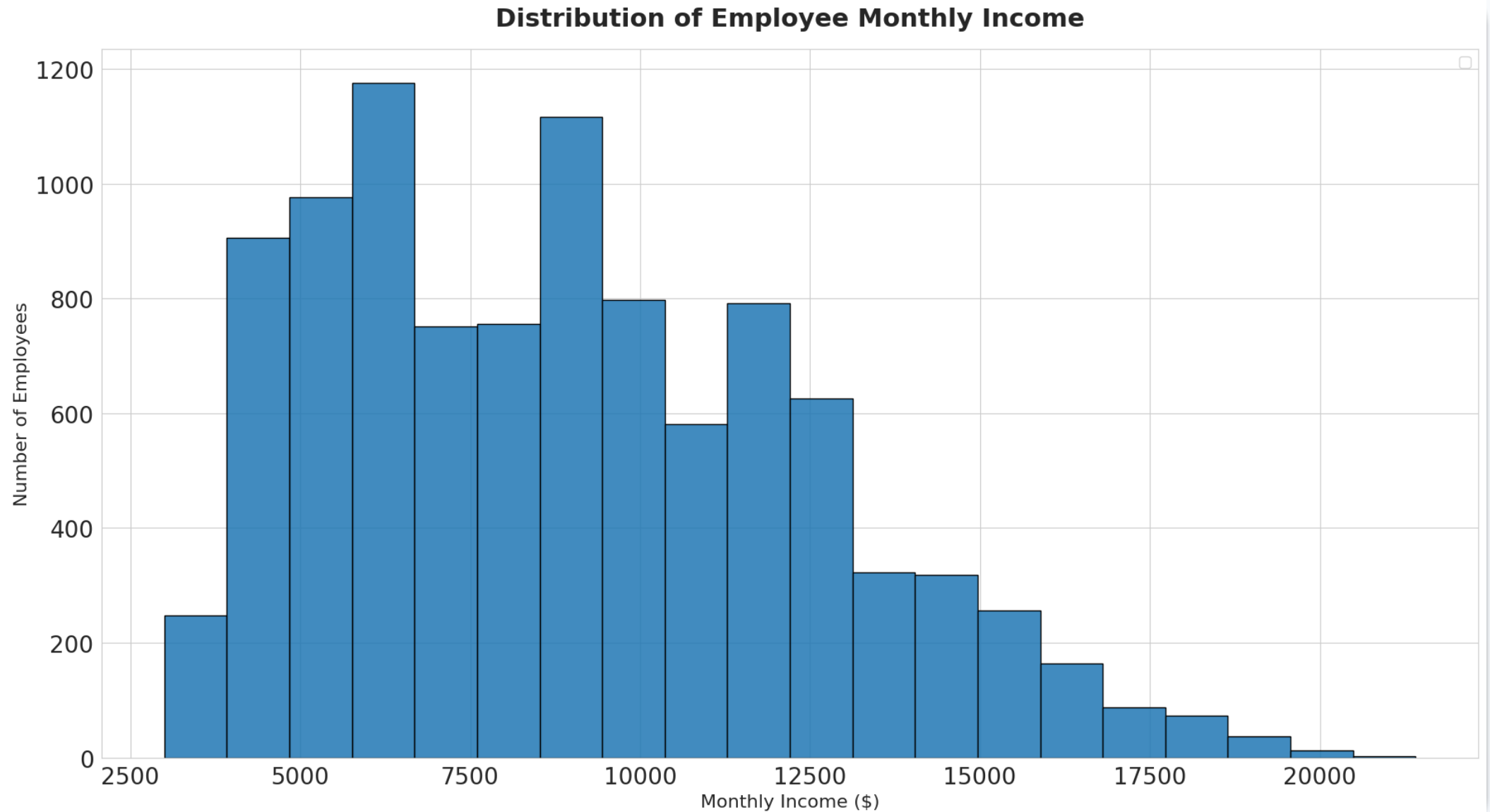


Exploratory Data Analysis



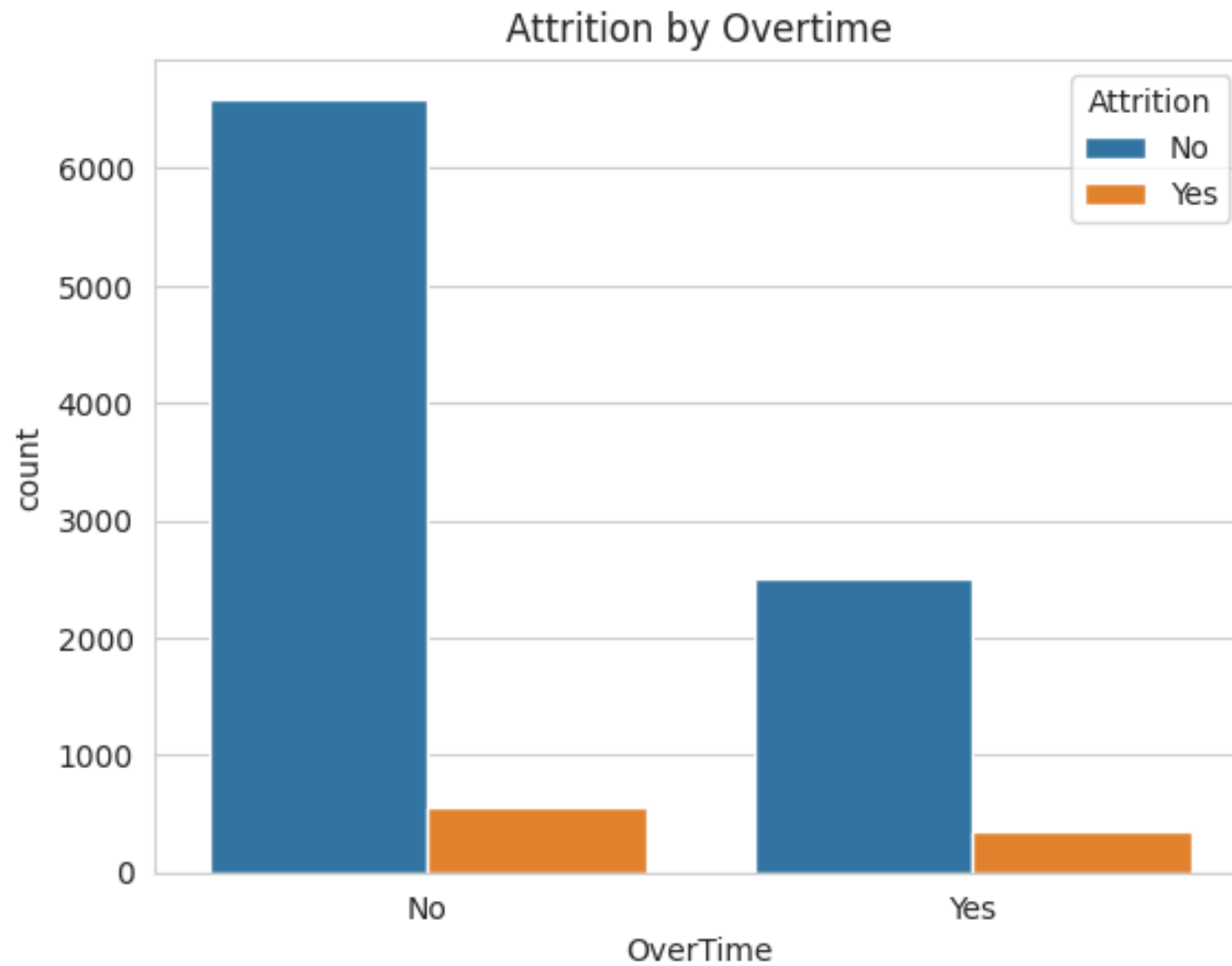


Exploratory Data Analysis



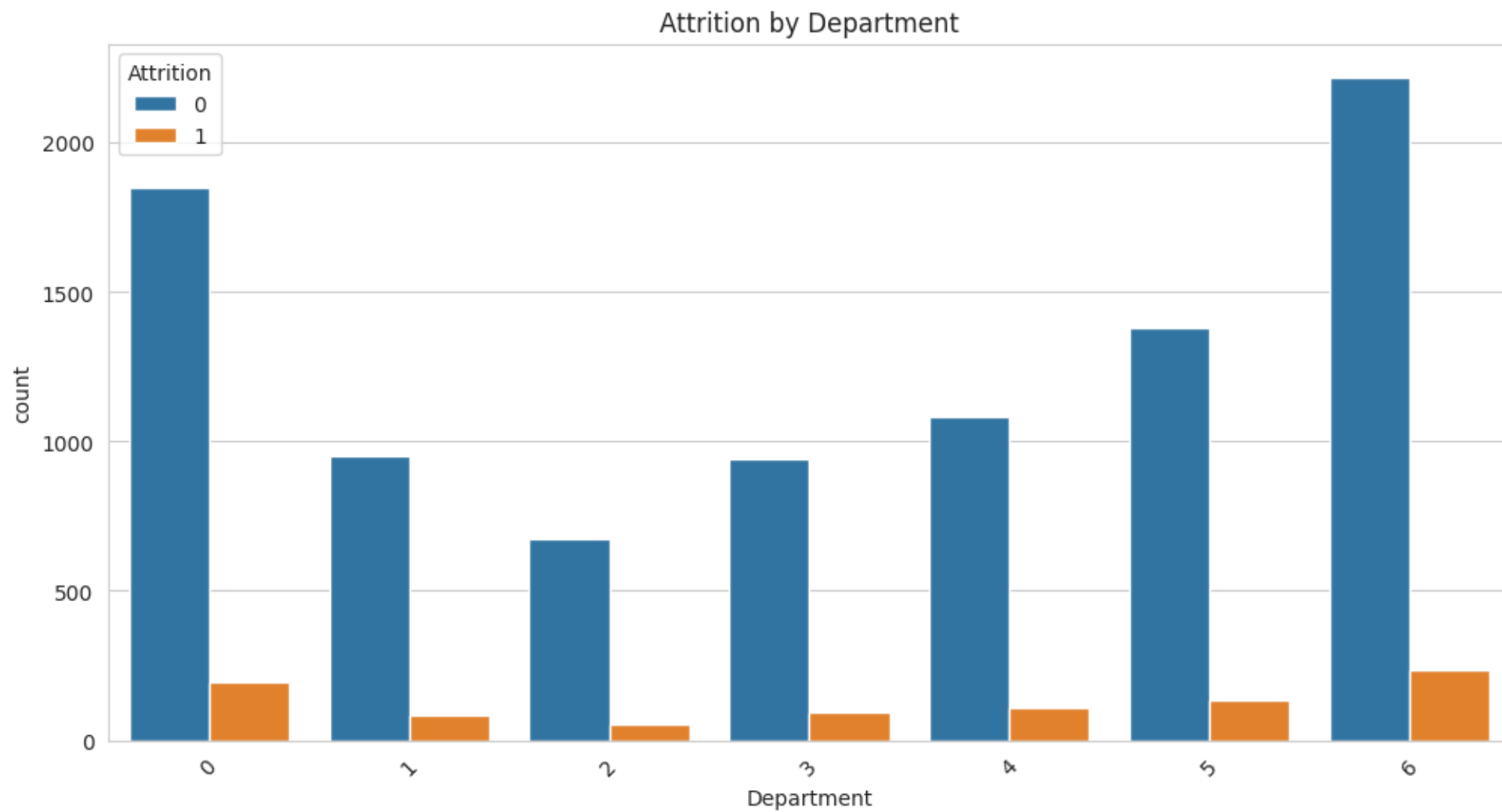


Exploratory Data Analysis



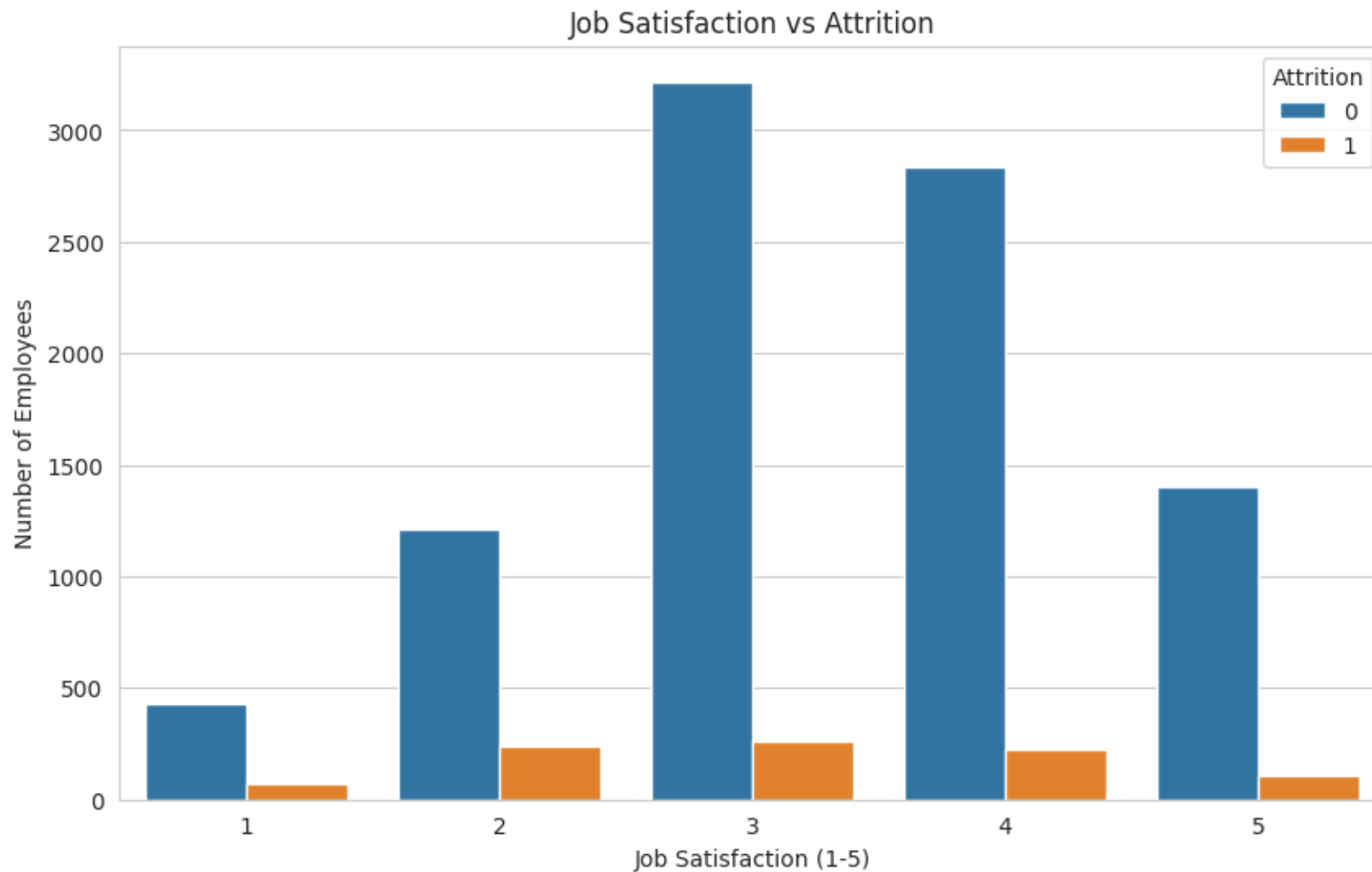


Exploratory Data Analysis



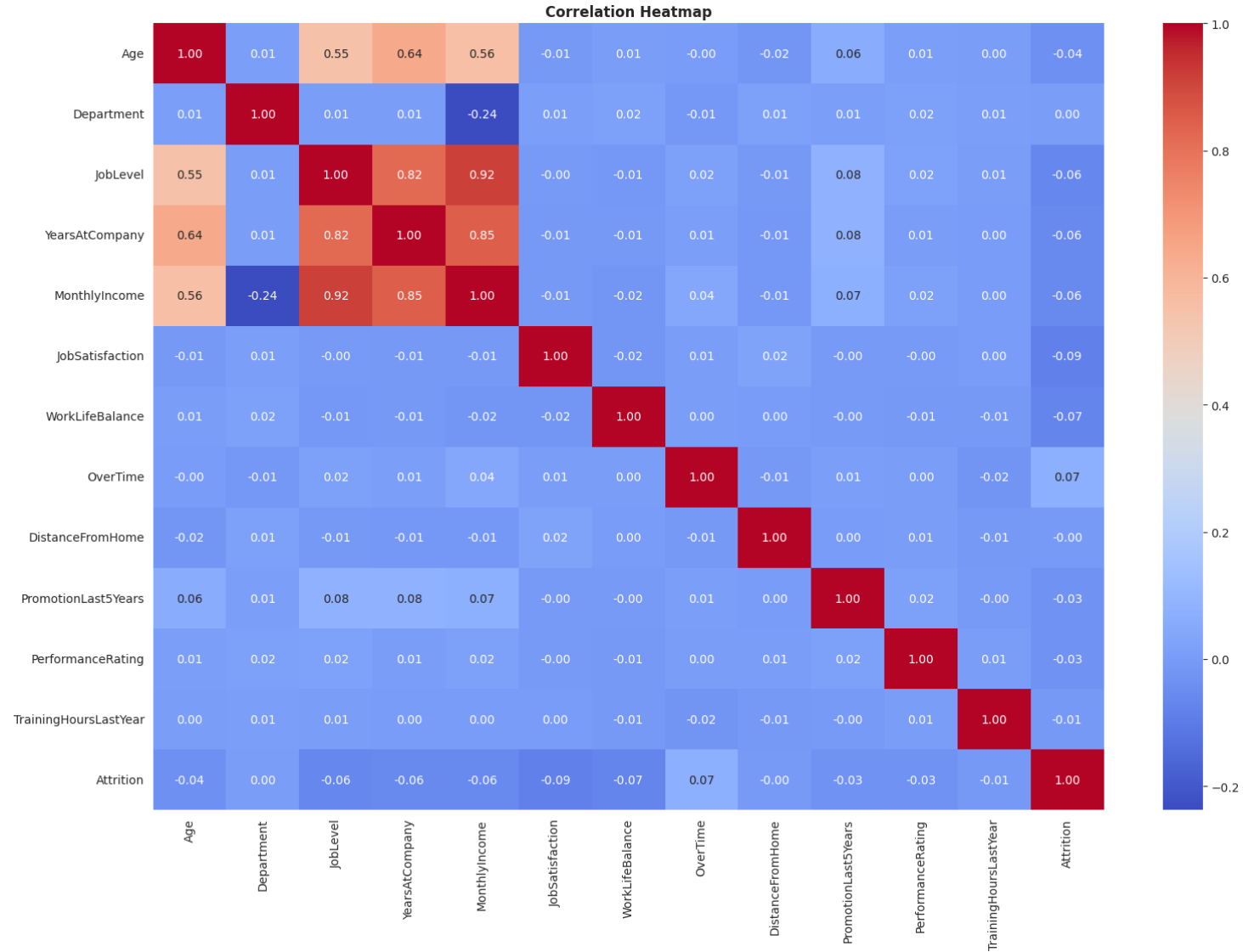


Exploratory Data Analysis





Exploratory Data Analysis





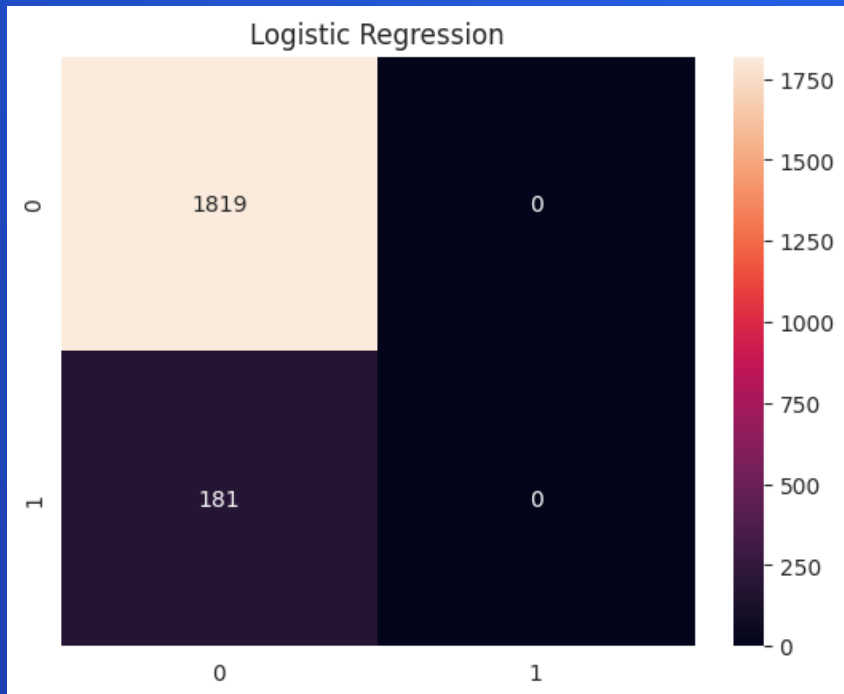
Machine Learning Model



Logistic Regression

Classifier Algorithm

Logistic Regression is well-suited for predicting whether an employee will stay or leave the organization.



Why Logistic Regression



High Accuracy

Robust predictions



Scalability

Handles large datasets



Reduces Overfitting

Ensemble approach



Feature Importance

simple, fast, and easy to interpret



Training Process



Train Model



Learn
Patterns



Predict
Attrition



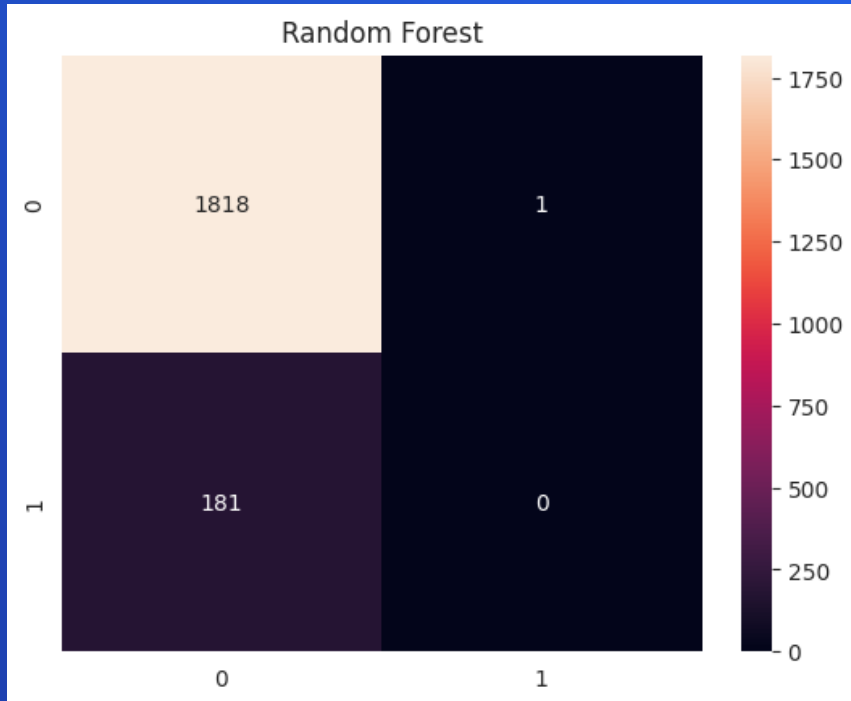
Machine Learning Model



Random Forest

Classifier Algorithm

An ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes for classification.



Why Random Forest?



High Accuracy

Robust predictions



Scalability

Handles large datasets



Reduces Overfitting

Ensemble approach



Feature Importance

Easy interpretation



Training Process



Train Model



Learn
Patterns



Predict
Attrition



Model Evaluation



Performance Metrics



Accuracy

Overall correctness



Precision

True positive rate



Recall

Sensitivity measure



F1 Score

Balanced metric



Confusion Matrix

Visualizes True Positives, True Negatives, False Positives, and False Negatives



Model Results



Accuracy

85%



Precision

83%



Recall

81%

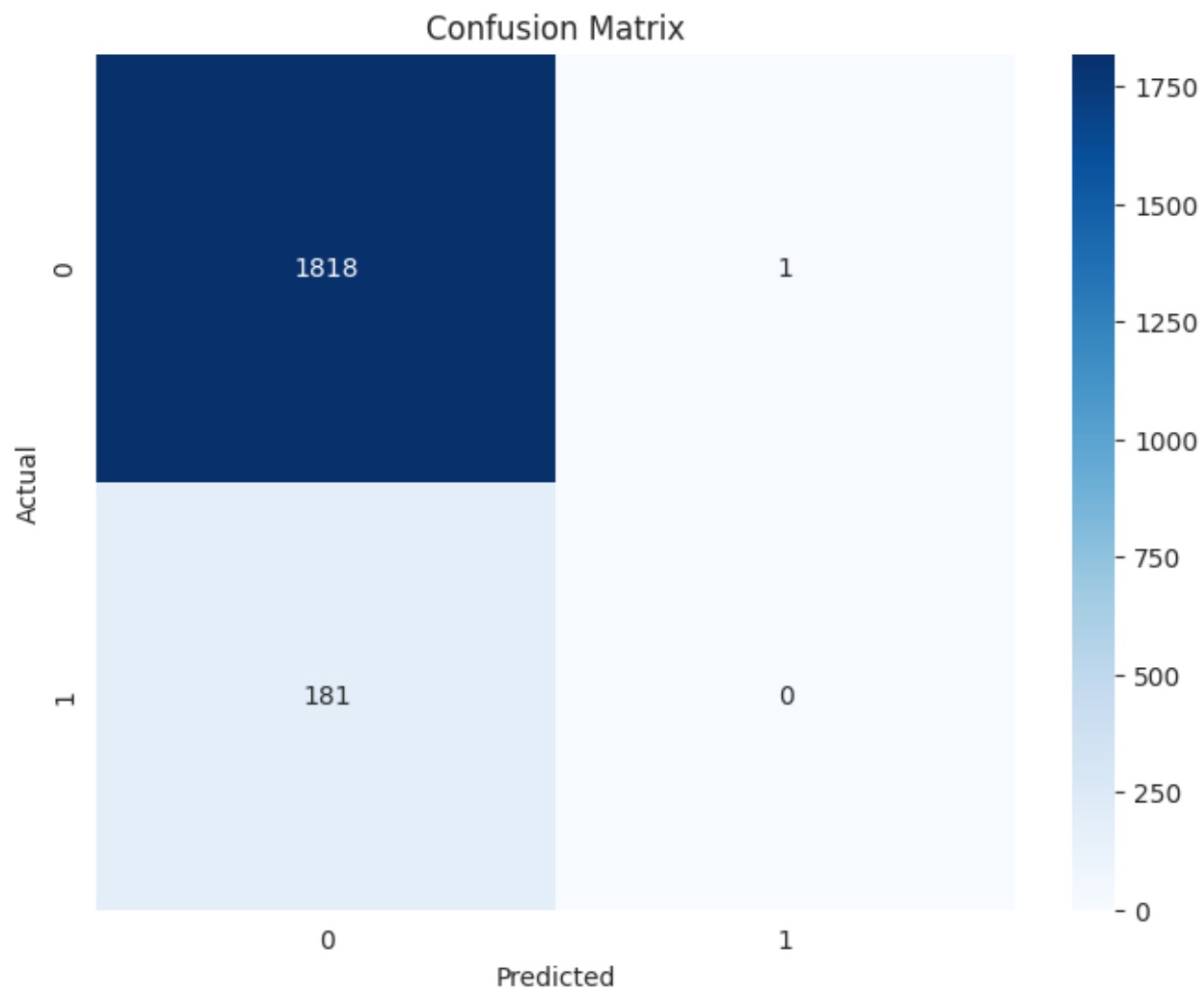


F1 Score

82%



Model Evaluation





Prediction System



Input



Feature

Age



Feature

Department



Feature

Salary



Feature

Experience



Feature

Satisfaction Level



ML Model

Prediction Engine



Output



Employee Likely to Stay

Retention strategies may not be urgent



Employee Likely to Leave

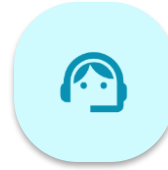
Intervention required immediately

★ Advantages



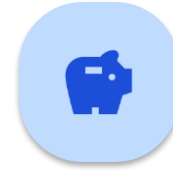
Reduces Turnover

Identifies at-risk employees early for proactive retention



Supports HR Decisions

Provides data-driven insights for strategic planning



Saves Recruitment Costs

Minimizes expenses associated with hiring and training



Improves Workforce Planning

Enables better resource allocation and succession planning



Increases Retention

Helps create a more stable and experienced workforce



Retain Talent

Employee attrition prediction helps organizations retain talent



Identify At-Risk Employees

Machine learning can identify employees at risk of leaving



HR Insights

The developed system provides valuable insights for HR departments



Data-Driven Decisions

The project supports data-driven decision making



Thank You



Questions?

Ready to answer your queries

